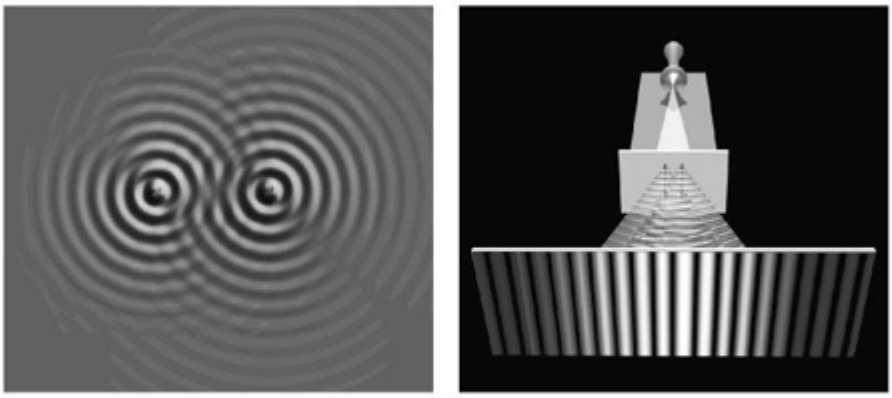


Figure 4.2 **(**a**)** Overlapping water waves produce an interference pattern. **(**b**)** Overlapping light waves produce an interference pattern.

In 1927, Clinton Davisson and Lester Germer fired a beam of electrons—particulate entities without any apparent connection to waves—at a piece of nickel crystal; the details need not concern us, but what does matter is that this experiment is equivalent to firing a beam of electrons at a barrier with two slits. When the experimenters allowed the electrons that passed through the slits to travel onward to a phosphor screen where their impact location was recorded by a tiny flash (the same kind of flashes responsible for the picture on your television screen), the results were astonishing. Thinking of the electrons as little pellets or bullets, you'd naturally expect their impact positions to line up with the two slits, as in Figure 4.3a. But that's not what Davisson and Germer found. Their experiment produced data schematically illustrated in Figure 4.3b: the electron impact positions filled out an interference pattern characteristic of waves. Davisson and Germer had found the smoking gun. *They had shown that the beam of particulate electrons must, unexpectedly, be some kind of wave.*

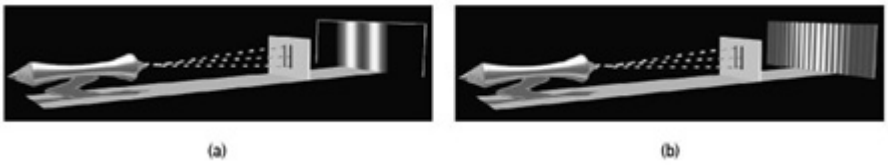


Figure 4.3 **(**a**)** Classical physics predicts that electrons fired at a barrier with two slits will produce two bright stripes on a detector. **(**b**)** Quantum physics predicts, and experiments confirm, that electrons will produce an interference pattern, showing that they embody wavelike features.